

Answer the following questions based on today's lesson.

Q1. What is enterprise software?

Enterprise software is software used by a **business or organization** to manage its daily work. It helps with things like **sales, employees, money, customers, and inventory**.

Example: A supermarket may use enterprise software to manage products, sales, and stock.

Q2. What does a software engineer need to think about besides writing code?

A software engineer needs to think about many things besides coding, such as:

- **Security:** Keeping the system and user information safe.
- **Performance:** Making sure the software works fast and smoothly.
- **Communication:** Making different software systems work together.
- **User needs:** Making the software easy and useful for users.
- **Reliability:** Making sure the software works correctly without frequent problems.
- **Maintenance:** Fixing and updating the software when needed.

Q3. How do different software systems communicate with each other?

Different software systems communicate using an **API (Application Programming Interface)**.

An API allows one software system to **request information or services from another system**.

Example: A shopping app can use a payment API to send payment information to a bank and receive a payment result.

Q4. How does software respond to events and protect users?

Software uses **events, authentication, and authorization**.

- **Event:** Something that happens in a system.
Example: A user clicks the "Buy Now" button.
- **Authentication:** Checking **who the user is**.
Example: Logging in with a username and password.
- **Authorization:** Checking **what the user is allowed to do**.
Example: A normal employee can view data but only a manager can delete it.

Q5. Why do software systems need standards and communication protocols?

Standards and protocols provide **common rules** for software systems.

They help different systems **understand each other and work together properly**.

Example: HTTP is a communication protocol used when a web browser communicates with a website.

In simple words, **standards and protocols are like common languages that allow different software systems to communicate**.

Q1. Identify six key business requirements of FreshGrocers.

FreshGrocers needs an ERP system to manage its main business activities.

- **Inventory Management:** Track products and stock levels in different stores.
 - **Sales and POS:** Record sales and process customer payments.
 - **CRM:** Manage customer information, preferences, and feedback.
 - **Supplier Management:** Manage suppliers, orders, and deliveries.
 - **Financial Management:** Manage sales, expenses, taxes, and financial reports.
 - **HR Management:** Manage employees, attendance, payroll, and recruitment.
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Q2. Recommend an ERP module for each FreshGrocers requirement.

- **Inventory Module:** Tracks stock and helps manage products.
 - **Sales/POS Module:** Records sales and processes payments.
 - **CRM Module:** Stores customer details and purchase history.
 - **Purchase/Supplier Module:** Manages suppliers and purchase orders.
 - **Accounting Module:** Manages money, expenses, taxes, and reports.
 - **HR Module:** Manages employee records, attendance, payroll, and recruitment.
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Q3. How can a customer purchase be processed through an integrated ERP system?

When a customer buys products, the **POS module** records the products and payment. The **Inventory module** automatically reduces the stock of the products sold. The **Accounting module** records the sales income. If the stock becomes low, the **Purchase module** can help order more products from suppliers.

Process:

Customer purchase → POS → Inventory update → Accounting → Reorder if needed

Q4. Give two benefits of ERP integration for FreshGrocers.

- **Saves time:** Information is updated automatically, so employees do not need to enter the same information many times.
 - **Better decisions:** Managers can easily see sales, stock, expenses, and other business information and make better decisions.
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Case Study 2: CleanGreen Laundry Services

Q5. Identify six key business requirements of CleanGreen.

CleanGreen needs an ERP system to manage its laundry operations.

- **Order Management:** Track orders from receiving to delivery.
 - **Inventory Management:** Manage detergents, cleaning supplies, and packaging.
 - **Scheduling and Delivery:** Schedule pickups and deliveries.
 - **CRM:** Manage customer details, preferences, and order history.
 - **Financial Management:** Manage payments, expenses, payroll, and reports.
 - **HR Management:** Manage employees, attendance, payroll, and recruitment.
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Q6. Recommend an ERP module for each CleanGreen requirement.

- **Order Management Module:** Tracks laundry orders from start to finish.
 - **Inventory Module:** Tracks detergents and other cleaning supplies.
 - **Planning/Delivery Module:** Schedules pickups and deliveries.
 - **CRM Module:** Stores customer information and order history.
 - **Accounting Module:** Manages payments, expenses, and financial reports.
 - **HR Module:** Manages employees, attendance, payroll, and recruitment.
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Q7. Explain the process of an online laundry order.

When a customer places an online order, the **Order Management/eCommerce module** records the order. The **Planning or Delivery module** schedules a pickup. After the clothes are collected, the order is sent for cleaning. The **Inventory module** keeps track of cleaning supplies used. The **Accounting module** records the payment. Finally, the **Delivery module** schedules delivery and the order is marked as completed.

Process:

Online order → Pickup → Cleaning → Payment → Delivery → Order completed

Q8. Give two benefits of ERP integration for CleanGreen.

- **Better order tracking:** The business can easily know whether an order is waiting, being cleaned, or ready for delivery.
 - **Better management:** Managers can see information about customers, employees, supplies, orders, and money in one system.
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Comparative Analysis

Q9. Compare FreshGrocers and CleanGreen.

Two common requirements:

1. **Inventory Management:** Both businesses need to track their supplies and stock.
2. **CRM:** Both need to store customer information and understand customer needs.

Two different requirements:

1. **POS:** FreshGrocers needs POS to process grocery sales in its stores.
 2. **Pickup and Delivery Scheduling:** CleanGreen needs scheduling to arrange laundry pickups and deliveries.
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Odoo ERP Exploration

Q10. Identify four Odoo applications that could support the businesses.

Odoo provides many applications for managing business activities. Four useful applications are:

1. Inventory

- **Business:** FreshGrocers and CleanGreen
- **Requirement:** Inventory management
- **Purpose:** Helps track products, supplies, stock levels, and replenishment.

2. Point of Sale (POS)

- **Business:** FreshGrocers
- **Requirement:** Sales management
- **Purpose:** Processes customer purchases and payments and helps update inventory.

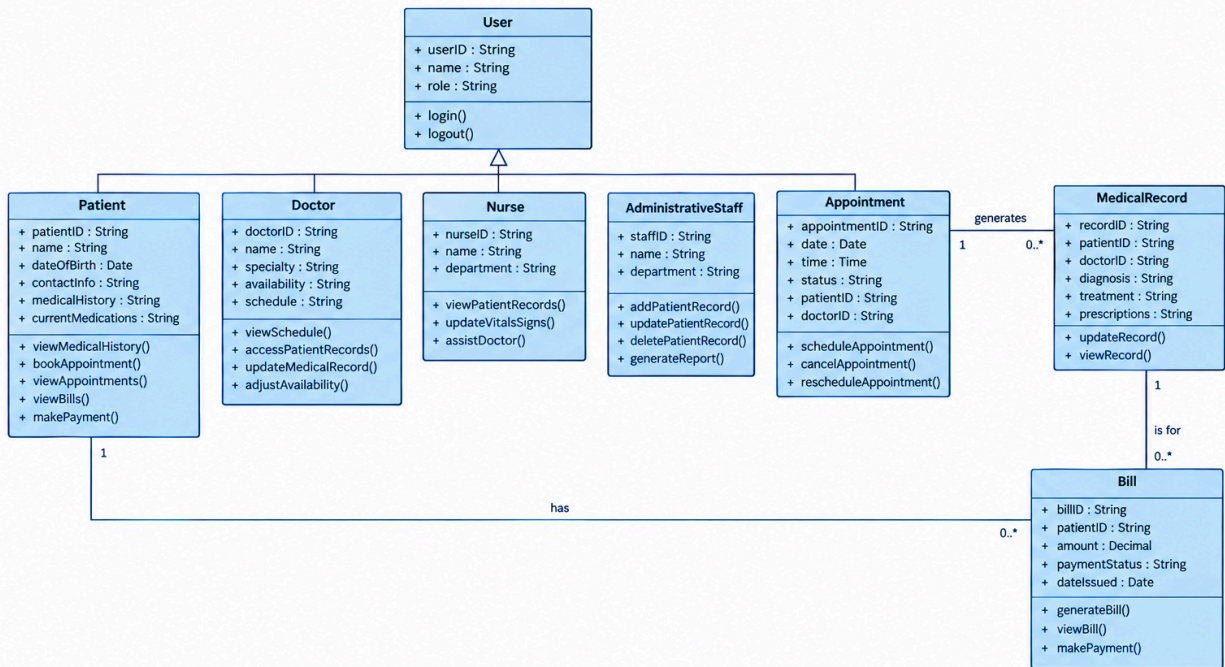
3. CRM

- **Business:** FreshGrocers and CleanGreen
- **Requirement:** Customer management
- **Purpose:** Stores customer information, purchase/order history, and customer interactions.

4. Accounting

- **Business:** FreshGrocers and CleanGreen
- **Requirement:** Financial management
- **Purpose:** Helps manage payments, sales, expenses, invoices, and financial records

UML Class Diagram – Hospital Management System (HMS)



UML Sequence Diagram – Booking an Appointment

